

Filter Summary Report: CG,TIA,Automated,simple,Z3,ZL

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Contents

1 Examined $H(z)$ for CG TIA Automated simple Z3 ZL: $\frac{Z_3 Z_L g_m}{Z_3 g_m + Z_L g_m}$

$$H(z) = \frac{Z_3 Z_L g_m}{Z_3 g_m + Z_L g_m}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(\infty, \infty, R_3, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{L_L R_3 s}{C_L L_L R_3 s^2 + L_L s + R_3}$$

Parameters:

Q: $\frac{\sqrt{C_L} R_3}{\sqrt{L_L}}$
 wo: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$
 bandwidth: $\frac{1}{C_L R_3}$
 K-LP: 0
 K-HP: 0
 K-BP: R_3
 Qz: None
 Wz: None

3.2 BP-2 $Z(s) = \left(\infty, \infty, R_3, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$

$$H(s) = \frac{L_L R_3 R_L s}{C_L L_L R_3 R_L s^2 + R_3 R_L + s (L_L R_3 + L_L R_L)}$$

Parameters:

Q: $\frac{\sqrt{C_L} R_3 R_L}{\sqrt{L_L} R_3 + \sqrt{L_L} R_L}$
 wo: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$
 bandwidth: $\frac{\sqrt{L_L} R_3 + \sqrt{L_L} R_L}{C_L \sqrt{L_L} R_3 R_L}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_3 R_L}{R_3 + R_L}$
 Qz: None
 Wz: None

3.3 BP-3 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$

$$H(s) = \frac{L_L R_L s}{L_L s + R_L + s^2 (C_3 L_L R_L + C_L L_L R_L)}$$

Parameters:

Q: $\frac{R_L \sqrt{C_3 + C_L}}{\sqrt{L_L}}$
 wo: $\frac{1}{\sqrt{C_3 L_L + C_L L_L}}$
 bandwidth: $\frac{\sqrt{L_L}}{R_L \sqrt{C_3 + C_L} \sqrt{C_3 L_L + C_L L_L}}$
 K-LP: 0
 K-HP: 0
 K-BP: R_L
 Qz: None
 Wz: None

3.4 BP-4 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

Parameters:

Q: $\frac{R_3 \sqrt{C_3 + C_L}}{\sqrt{L_L}}$
 wo: $\frac{1}{\sqrt{C_3 L_L + C_L L_L}}$
 bandwidth: $\frac{\sqrt{L_L}}{R_3 \sqrt{C_3 + C_L} \sqrt{C_3 L_L + C_L L_L}}$
 K-LP: 0
 K-HP: 0
 K-BP: R_3
 Qz: None
 Wz: None

$$H(s) = \frac{L_L R_3 s}{L_L s + R_3 + s^2 (C_3 L_L R_3 + C_L L_L R_3)}$$

3.5 BP-5 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$

Parameters:

Q: $\frac{R_3 R_L \sqrt{C_3 + C_L}}{\sqrt{L_L} R_3 + \sqrt{L_L} R_L}$
 wo: $\frac{1}{\sqrt{C_3 L_L + C_L L_L}}$
 bandwidth: $\frac{\sqrt{L_L} R_3 + \sqrt{L_L} R_L}{R_3 R_L \sqrt{C_3 + C_L} \sqrt{C_3 L_L + C_L L_L}}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_3 R_L}{R_3 + R_L}$
 Qz: None
 Wz: None

$$H(s) = \frac{L_L R_3 R_L s}{R_3 R_L + s^2 (C_3 L_L R_3 R_L + C_L L_L R_3 R_L) + s (L_L R_3 + L_L R_L)}$$

3.6 BP-6 $Z(s) = \left(\infty, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \infty, \infty, R_L \right)$

Parameters:

Q: $\frac{\sqrt{C_3} R_L}{\sqrt{L_3}}$
 wo: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$
 bandwidth: $\frac{1}{C_3 R_L}$
 K-LP: 0
 K-HP: 0
 K-BP: R_L
 Qz: None
 Wz: None

$$H(s) = \frac{L_3 R_L s}{C_3 L_3 R_L s^2 + L_3 s + R_L}$$

3.7 BP-7 $Z(s) = \left(\infty, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

Parameters:

Q: $\frac{R_L \sqrt{C_3 + C_L}}{\sqrt{L_3}}$
 wo: $\frac{1}{\sqrt{C_3 L_3 + C_L L_3}}$
 bandwidth: $\frac{\sqrt{L_3}}{R_L \sqrt{C_3 + C_L} \sqrt{C_3 L_3 + C_L L_3}}$
 K-LP: 0

$$H(s) = \frac{L_3 R_L s}{L_3 s + R_L + s^2 (C_3 L_3 R_L + C_L L_3 R_L)}$$

K-HP: 0
K-BP: R_L
Qz: None
Wz: None

$$\mathbf{3.8 \quad BP-8} \quad Z(s) = \left(\infty, \infty, \frac{L_3 s}{C_3 L_3 s^2 + 1}, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$$

Parameters:

Q: $\frac{R_L \sqrt{C_3 + C_L} \sqrt{L_3 + L_L}}{\sqrt{L_3} \sqrt{L_L}}$
wo: $\frac{\sqrt{L_3 + L_L}}{\sqrt{C_3 L_3 L_L + C_L L_3 L_L}}$
bandwidth: $\frac{\sqrt{L_3} \sqrt{L_L}}{R_L \sqrt{C_3 + C_L} \sqrt{C_3 L_3 L_L + C_L L_3 L_L}}$
K-LP: 0
K-HP: 0
K-BP: R_L
Qz: None
Wz: None

$$H(s) = \frac{L_3 L_L R_L s}{L_3 L_L s + L_3 R_L + L_L R_L + s^2 (C_3 L_3 L_L R_L + C_L L_3 L_L R_L)}$$

$$\mathbf{3.9 \quad BP-9} \quad Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, R_L \right)$$

Parameters:

Q: $\frac{\sqrt{C_3} R_3 R_L}{\sqrt{L_3} R_3 + \sqrt{L_3} R_L}$
wo: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$
bandwidth: $\frac{\sqrt{L_3} R_3 + \sqrt{L_3} R_L}{C_3 \sqrt{L_3} R_3 R_L}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_3 R_L}{R_3 + R_L}$
Qz: None
Wz: None

$$H(s) = \frac{L_3 R_3 R_L s}{C_3 L_3 R_3 R_L s^2 + R_3 R_L + s (L_3 R_3 + L_3 R_L)}$$

$$\mathbf{3.10 \quad BP-10} \quad Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, \frac{1}{C_L s} \right)$$

Parameters:

Q: $\frac{R_3 \sqrt{C_3 + C_L}}{\sqrt{L_3} 1}$
wo: $\frac{1}{\sqrt{C_3 L_3 + C_L L_3}}$
bandwidth: $\frac{\sqrt{L_3}}{R_3 \sqrt{C_3 + C_L} \sqrt{C_3 L_3 + C_L L_3}}$
K-LP: 0
K-HP: 0
K-BP: R_3
Qz: None
Wz: None

$$H(s) = \frac{L_3 R_3 s}{L_3 s + R_3 + s^2 (C_3 L_3 R_3 + C_L L_3 R_3)}$$

3.11 BP-11 $Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{L_3 R_3 R_L s}{R_3 R_L + s^2 (C_3 L_3 R_3 R_L + C_L L_3 R_3 R_L) + s (L_3 R_3 + L_3 R_L)}$$

Parameters:

Q: $\frac{R_3 R_L \sqrt{C_3 + C_L}}{\sqrt{L_3 R_3 + \sqrt{L_3 R_L}}}$
 wo: $\frac{1}{\sqrt{C_3 L_3 + C_L L_3}}$
 bandwidth: $\frac{\sqrt{L_3 R_3 + \sqrt{L_3 R_L}}}{R_3 R_L \sqrt{C_3 + C_L} \sqrt{C_3 L_3 + C_L L_3}}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_3 R_L}{R_3 + R_L}$
 Qz: None
 Wz: None

3.12 BP-12 $Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{L_3 L_L R_3 s}{L_3 L_L s + L_3 R_3 + L_L R_3 + s^2 (C_3 L_3 L_L R_3 + C_L L_3 L_L R_3)}$$

Parameters:

Q: $\frac{R_3 \sqrt{C_3 + C_L} \sqrt{L_3 + L_L}}{\sqrt{L_3} \sqrt{L_L}}$
 wo: $\frac{\sqrt{L_3 + L_L}}{\sqrt{C_3 L_3 L_L + C_L L_3 L_L}}$
 bandwidth: $\frac{\sqrt{L_3} \sqrt{L_L}}{R_3 \sqrt{C_3 + C_L} \sqrt{C_3 L_3 L_L + C_L L_3 L_L}}$
 K-LP: 0
 K-HP: 0
 K-BP: R_3
 Qz: None
 Wz: None

3.13 BP-13 $Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$

$$H(s) = \frac{L_3 L_L R_3 R_L s}{L_3 R_3 R_L + L_L R_3 R_L + s^2 (C_3 L_3 L_L R_3 R_L + C_L L_3 L_L R_3 R_L) + s (L_3 L_L R_3 + L_3 L_L R_L)}$$

Parameters:

Q: $\frac{R_3 R_L \sqrt{C_3 + C_L} \sqrt{L_3 + L_L}}{\sqrt{L_3} \sqrt{L_L} R_3 + \sqrt{L_3} \sqrt{L_L} R_L}$
 wo: $\frac{\sqrt{L_3 + L_L}}{\sqrt{C_3 L_3 L_L + C_L L_3 L_L}}$
 bandwidth: $\frac{\sqrt{L_3} \sqrt{L_L} R_3 + \sqrt{L_3} \sqrt{L_L} R_L}{R_3 R_L \sqrt{C_3 + C_L} \sqrt{C_3 L_3 L_L + C_L L_3 L_L}}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_3 R_L}{R_3 + R_L}$
 Qz: None
 Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(\infty, \infty, R_3, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_L L_L R_3 s^2 + R_3}{C_L L_L s^2 + C_L R_3 s + 1}$$

Parameters:

Q: $\frac{\sqrt{L_L}}{\sqrt{C_L R_3}}$
 wo: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$
 bandwidth: $\frac{R_3}{L_L}$
 K-LP: R_3
 K-HP: R_3
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$

$$4.2 \quad \text{BS-2} \quad Z(s) = \left(\infty, \infty, R_3, \infty, \infty, \frac{R_L(C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_L L_L R_3 R_L s^2 + R_3 R_L}{C_L R_3 R_L s + R_3 + R_L + s^2 (C_L L_L R_3 + C_L L_L R_L)}$$

Parameters:

Q: $\frac{\sqrt{L_L} R_3 + \sqrt{L_L} R_L}{\sqrt{C_L} R_3 R_L}$
 wo: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$
 bandwidth: $\frac{R_3 R_L}{\sqrt{L_L} (\sqrt{L_L} R_3 + \sqrt{L_L} R_L)}$
 K-LP: $\frac{R_3 R_L}{R_3 + R_L}$
 K-HP: $\frac{R_3 R_L}{R_3 + R_L}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$

$$4.3 \quad \text{BS-3} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \infty, \infty, R_L \right)$$

$$H(s) = \frac{C_3 L_3 R_L s^2 + R_L}{C_3 L_3 s^2 + C_3 R_L s + 1}$$

Parameters:

Q: $\frac{\sqrt{L_3}}{\sqrt{C_3} R_L}$
 wo: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$
 bandwidth: $\frac{R_L}{L_3}$
 K-LP: R_L
 K-HP: R_L
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$

$$4.4 \quad \text{BS-4} \quad Z(s) = \left(\infty, \infty, \frac{R_3(C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \infty, \infty, R_L \right)$$

$$H(s) = \frac{C_3 L_3 R_3 R_L s^2 + R_3 R_L}{C_3 R_3 R_L s + R_3 + R_L + s^2 (C_3 L_3 R_3 + C_3 L_3 R_L)}$$

Parameters:

Q: $\frac{\sqrt{L_3} R_3 + \sqrt{L_3} R_L}{\sqrt{C_3} R_3 R_L}$
 wo: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$
 bandwidth: $\frac{R_3 R_L}{\sqrt{L_3} (\sqrt{L_3} R_3 + \sqrt{L_3} R_L)}$
 K-LP: $\frac{R_3 R_L}{R_3 + R_L}$
 K-HP: $\frac{R_3 R_L}{R_3 + R_L}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_3} \sqrt{L_3}}$

5 GE

5.1 GE-1 $Z(s) = \left(\infty, \infty, R_3, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{L_L}}{\sqrt{C_L R_3} + \sqrt{C_L R_L}} \\ \text{wo: } & \frac{1}{\sqrt{C_L} \sqrt{L_L}} \\ \text{bandwidth: } & \frac{\sqrt{C_L R_3} + \sqrt{C_L R_L}}{\sqrt{C_L L_L}} \\ \text{K-LP: } & R_3 \\ \text{K-HP: } & R_3 \\ \text{K-BP: } & \frac{R_3 R_L}{R_3 + R_L} \\ \text{Qz: } & \frac{\sqrt{L_L}}{\sqrt{C_L R_L}} \\ \text{Wz: } & \frac{1}{\sqrt{C_L} \sqrt{L_L}} \end{aligned}$$

5.2 GE-2 $Z(s) = \left(\infty, \infty, R_3, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_L R_3} + \sqrt{C_L R_L}}{\sqrt{L_L}} \\ \text{wo: } & \frac{1}{\sqrt{C_L} \sqrt{L_L}} \\ \text{bandwidth: } & \frac{1}{\sqrt{C_L} (\sqrt{C_L R_3} + \sqrt{C_L R_L})} \\ \text{K-LP: } & \frac{R_3 R_L}{R_3 + R_L} \\ \text{K-HP: } & \frac{R_3 R_L}{R_3 + R_L} \\ \text{K-BP: } & R_3 \\ \text{Qz: } & \frac{\sqrt{C_L R_L}}{\sqrt{L_L}} \\ \text{Wz: } & \frac{1}{\sqrt{C_L} \sqrt{L_L}} \end{aligned}$$

5.3 GE-3 $Z(s) = \left(\infty, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L \right)$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{L_3}}{\sqrt{C_3 R_3} + \sqrt{C_3 R_L}} \\ \text{wo: } & \frac{1}{\sqrt{C_3} \sqrt{L_3}} \\ \text{bandwidth: } & \frac{\sqrt{C_3 R_3} + \sqrt{C_3 R_L}}{\sqrt{C_3 L_3}} \\ \text{K-LP: } & R_L \\ \text{K-HP: } & R_L \\ \text{K-BP: } & \frac{R_3 R_L}{R_3 + R_L} \\ \text{Qz: } & \frac{\sqrt{L_3}}{\sqrt{C_3 R_3}} \\ \text{Wz: } & \frac{1}{\sqrt{C_3} \sqrt{L_3}} \end{aligned}$$

$$H(s) = \frac{C_L L_L R_3 s^2 + C_L R_3 R_L s + R_3}{C_L L_L s^2 + s (C_L R_3 + C_L R_L) + 1}$$

$$H(s) = \frac{C_L L_L R_3 R_L s^2 + L_L R_3 s + R_3 R_L}{L_L s + R_3 + R_L + s^2 (C_L L_L R_3 + C_L L_L R_L)}$$

$$H(s) = \frac{C_3 L_3 R_L s^2 + C_3 R_3 R_L s + R_L}{C_3 L_3 s^2 + s (C_3 R_3 + C_3 R_L) + 1}$$

5.4 GE-4 $Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \infty, \infty, R_L \right)$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_3} R_3 + \sqrt{C_3} R_L}{\sqrt{L_3}} \\ \text{wo: } & \frac{1}{\sqrt{C_3} \sqrt{L_3}} \\ \text{bandwidth: } & \frac{1}{\sqrt{C_3} (\sqrt{C_3} R_3 + \sqrt{C_3} R_L)} \\ \text{K-LP: } & \frac{R_3 R_L}{R_3 + R_L} \\ \text{K-HP: } & \frac{R_3 R_L}{R_3 + R_L} \\ \text{K-BP: } & \frac{R_L}{R_3 + R_L} \\ \text{Qz: } & \frac{\sqrt{C_3} R_3}{\sqrt{L_3}} \\ \text{Wz: } & \frac{1}{\sqrt{C_3} \sqrt{L_3}} \end{aligned}$$

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_L R_3 R_L s + R_3}{C_3 C_L R_3 R_L s^2 + s (C_3 R_3 + C_L R_3 + C_L R_L) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_3} \sqrt{C_L} \sqrt{R_3} \sqrt{R_L}}{C_3 R_3 + C_L R_3 + C_L R_L} \\ \text{wo: } & \frac{1}{\sqrt{C_3} \sqrt{C_L} \sqrt{R_3} \sqrt{R_L}} \\ \text{bandwidth: } & \frac{C_3 R_3 + C_L R_3 + C_L R_L}{C_3 C_L R_3 R_L} \\ \text{K-LP: } & R_3 \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_L R_3 R_L}{C_3 R_3 + C_L R_3 + C_L R_L} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{C_3 R_3 R_L s + R_L}{C_3 C_L R_3 R_L s^2 + s (C_3 R_3 + C_3 R_L + C_L R_L) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_3} \sqrt{C_L} \sqrt{R_3} \sqrt{R_L}}{C_3 R_3 + C_3 R_L + C_L R_L} \\ \text{wo: } & \frac{1}{\sqrt{C_3} \sqrt{C_L} \sqrt{R_3} \sqrt{R_L}} \\ \text{bandwidth: } & \frac{C_3 R_3 + C_3 R_L + C_L R_L}{C_3 C_L R_3 R_L} \\ \text{K-LP: } & R_L \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_3 R_3 R_L}{C_3 R_3 + C_3 R_L + C_L R_L} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty, \infty, R_3, \infty, \infty, R_L)$

$$H(s) = \frac{R_3 R_L}{R_3 + R_L}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty, \infty, R_3, \infty, \infty, \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3}{C_L R_3 s + 1}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty, \infty, R_3, \infty, \infty, \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_3 R_L}{C_L R_3 R_L s + R_3 + R_L}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty, \infty, R_3, \infty, \infty, R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_3 R_L s + R_3}{s(C_L R_3 + C_L R_L) + 1}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, R_L\right)$

$$H(s) = \frac{R_L}{C_3 R_L s + 1}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{1}{C_L s}\right)$

$$H(s) = \frac{1}{s(C_3 + C_L)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_L}{s(C_3 R_L + C_L R_L) + 1}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_L s + 1}{C_3 C_L R_L s^2 + s(C_3 + C_L)}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L L_L s^2 + 1}{C_3 C_L L_L s^3 + s(C_3 + C_L)}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{L_L s}{s^2 (C_3 L_L + C_L L_L) + 1}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L L_L s^2 + C_L R_L s + 1}{C_3 C_L L_L s^3 + C_3 C_L R_L s^2 + s (C_3 + C_L)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_3 C_L L_L R_L s^3 + C_3 R_L s + s^2 (C_3 L_L + C_L L_L) + 1}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{R_L (C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_L L_L R_L s^2 + R_L}{C_3 C_L L_L R_L s^3 + C_L L_L s^2 + s (C_3 R_L + C_L R_L) + 1}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, R_L \right)$$

$$H(s) = \frac{R_3 R_L}{C_3 R_3 R_L s + R_3 + R_L}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_3}{s (C_3 R_3 + C_L R_3) + 1}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_3 R_L}{R_3 + R_L + s (C_3 R_3 R_L + C_L R_3 R_L)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L L_L R_3 s^2 + R_3}{C_3 C_L L_L R_3 s^3 + C_L L_L s^2 + s (C_3 R_3 + C_L R_3) + 1}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L L_L R_3 s^2 + C_L R_3 R_L s + R_3}{C_3 C_L L_L R_3 s^3 + s^2 (C_3 C_L R_3 R_L + C_L L_L) + s (C_3 R_3 + C_L R_3 + C_L R_L) + 1}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_L L_L R_3 R_L s^2 + L_L R_3 s + R_3 R_L}{C_3 C_L L_L R_3 R_L s^3 + R_3 + R_L + s^2 (C_3 L_L R_3 + C_L L_L R_3 + C_L L_L R_L) + s (C_3 R_3 R_L + L_L)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{R_L(C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_L L_L R_3 R_L s^2 + R_3 R_L}{C_3 C_L L_L R_3 R_L s^3 + R_3 + R_L + s^2 (C_L L_L R_3 + C_L L_L R_L) + s (C_3 R_3 R_L + C_L R_3 R_L)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L \right)$$

$$H(s) = \frac{C_3 R_3 R_L s + R_L}{s (C_3 R_3 + C_3 R_L) + 1}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 R_3 s + 1}{C_3 C_L R_3 s^2 + s (C_3 + C_L)}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L R_3 R_L s^2 + s (C_3 R_3 + C_L R_L) + 1}{s^2 (C_3 C_L R_3 + C_3 C_L R_L) + s (C_3 + C_L)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L L_L R_3 s^3 + C_3 R_3 s + C_L L_L s^2 + 1}{C_3 C_L L_L s^3 + C_3 C_L R_3 s^2 + s (C_3 + C_L)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_3 L_L R_3 s^2 + L_L s}{C_3 C_L L_L R_3 s^3 + C_3 R_3 s + s^2 (C_3 L_L + C_L L_L) + 1}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L L_L R_3 s^3 + s^2 (C_3 C_L R_3 R_L + C_L L_L) + s (C_3 R_3 + C_L R_L) + 1}{C_3 C_L L_L s^3 + s^2 (C_3 C_L R_3 + C_3 C_L R_L) + s (C_3 + C_L)}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$$

$$H(s) = \frac{C_3 L_L R_3 R_L s^2 + L_L R_L s}{C_3 C_L L_L R_3 R_L s^3 + R_L + s^2 (C_3 L_L R_3 + C_3 L_L R_L + C_L L_L R_L) + s (C_3 R_3 R_L + L_L)}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_3 C_L L_L R_3 R_L s^3 + R_L + s^2 (C_3 L_L R_3 + C_L L_L R_L) + s (C_3 R_3 R_L + L_L)}{s^3 (C_3 C_L L_L R_3 + C_3 C_L L_L R_L) + s^2 (C_3 L_L + C_L L_L) + s (C_3 R_3 + C_3 R_L) + 1}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{R_L(C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_3 C_L L_L R_3 R_L s^3 + C_3 R_3 R_L s + C_L L_L R_L s^2 + R_L}{s^3 (C_3 C_L L_L R_3 + C_3 C_L L_L R_L) + s^2 (C_3 C_L R_3 R_L + C_L L_L) + s (C_3 R_3 + C_3 R_L + C_L R_L) + 1}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 L_3 s^2 + 1}{C_3 C_L L_3 s^3 + s (C_3 + C_L)}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_3 L_3 R_L s^2 + R_L}{C_3 C_L L_3 R_L s^3 + C_3 L_3 s^2 + s (C_3 R_L + C_L R_L) + 1}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L L_3 R_L s^3 + C_3 L_3 s^2 + C_L R_L s + 1}{C_3 C_L L_3 s^3 + C_3 C_L R_L s^2 + s (C_3 + C_L)}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L L_3 L_L s^4 + s^2 (C_3 L_3 + C_L L_L) + 1}{s^3 (C_3 C_L L_3 + C_3 C_L L_L) + s (C_3 + C_L)}$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_3 L_3 L_L s^3 + L_L s}{C_3 C_L L_3 L_L s^4 + s^2 (C_3 L_3 + C_3 L_L + C_L L_L) + 1}$$

$$\mathbf{9.35 \quad X-INVALID-ORDER-35} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L L_3 L_L s^4 + C_3 C_L L_3 R_L s^3 + C_L R_L s + s^2 (C_3 L_3 + C_L L_L) + 1}{C_3 C_L R_L s^2 + s^3 (C_3 C_L L_3 + C_3 C_L L_L) + s (C_3 + C_L)}$$

$$\mathbf{9.36 \quad X-INVALID-ORDER-36} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$$

$$H(s) = \frac{C_3 L_3 L_L R_L s^3 + L_L R_L s}{C_3 C_L L_3 L_L R_L s^4 + C_3 L_3 L_L s^3 + L_L s + R_L + s^2 (C_3 L_3 R_L + C_3 L_L R_L + C_L L_L R_L)}$$

$$\mathbf{9.37 \quad X-INVALID-ORDER-37} \quad Z(s) = \left(\infty, \infty, L_3 s + \frac{1}{C_3 s}, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_3 C_L L_3 L_L R_L s^4 + C_3 L_3 L_L s^3 + L_L s + R_L + s^2 (C_3 L_3 R_L + C_L L_L R_L)}{C_3 C_L L_3 L_L s^4 + C_3 C_L L_L R_L s^3 + C_3 R_L s + s^2 (C_3 L_3 + C_3 L_L + C_L L_L) + 1}$$

$$\mathbf{9.38 \quad X-INVALID-ORDER-38} \quad Z(s) = \left(\infty, \infty, L_3s + \frac{1}{C_3s}, \infty, \infty, \frac{R_L(C_LL_Ls^2+1)}{C_LL_Ls^2+C_LR_Ls+1} \right)$$

$$H(s) = \frac{C_3C_LL_3L_LR_Ls^4 + R_L + s^2(C_3L_3R_L + C_LL_LR_L)}{C_3C_LL_3L_Ls^4 + s^3(C_3C_LL_3R_L + C_3C_LL_LR_L) + s^2(C_3L_3 + C_LL_L) + s(C_3R_L + C_LR_L) + 1}$$

$$\mathbf{9.39 \quad X-INVALID-ORDER-39} \quad Z(s) = \left(\infty, \infty, \frac{L_3s}{C_3L_3s^2+1}, \infty, \infty, \frac{1}{C_Ls} \right)$$

$$H(s) = \frac{L_3s}{s^2(C_3L_3 + C_LL_3) + 1}$$

$$\mathbf{9.40 \quad X-INVALID-ORDER-40} \quad Z(s) = \left(\infty, \infty, \frac{L_3s}{C_3L_3s^2+1}, \infty, \infty, R_L + \frac{1}{C_Ls} \right)$$

$$H(s) = \frac{C_LL_3R_Ls^2 + L_3s}{C_3C_LL_3R_Ls^3 + C_LR_Ls + s^2(C_3L_3 + C_LL_3) + 1}$$

$$\mathbf{9.41 \quad X-INVALID-ORDER-41} \quad Z(s) = \left(\infty, \infty, \frac{L_3s}{C_3L_3s^2+1}, \infty, \infty, L_Ls + \frac{1}{C_Ls} \right)$$

$$H(s) = \frac{C_LL_3L_Ls^3 + L_3s}{C_3C_LL_3L_Ls^4 + s^2(C_3L_3 + C_LL_3 + C_LL_L) + 1}$$

$$\mathbf{9.42 \quad X-INVALID-ORDER-42} \quad Z(s) = \left(\infty, \infty, \frac{L_3s}{C_3L_3s^2+1}, \infty, \infty, \frac{L_Ls}{C_LL_Ls^2+1} \right)$$

$$H(s) = \frac{L_3L_Ls}{L_3 + L_L + s^2(C_3L_3L_L + C_LL_3L_L)}$$

$$\mathbf{9.43 \quad X-INVALID-ORDER-43} \quad Z(s) = \left(\infty, \infty, \frac{L_3s}{C_3L_3s^2+1}, \infty, \infty, L_Ls + R_L + \frac{1}{C_Ls} \right)$$

$$H(s) = \frac{C_LL_3L_Ls^3 + C_LL_3R_Ls^2 + L_3s}{C_3C_LL_3L_Ls^4 + C_3C_LL_3R_Ls^3 + C_LR_Ls + s^2(C_3L_3 + C_LL_3 + C_LL_L) + 1}$$

$$\mathbf{9.44 \quad X-INVALID-ORDER-44} \quad Z(s) = \left(\infty, \infty, \frac{L_3s}{C_3L_3s^2+1}, \infty, \infty, \frac{C_LL_LR_Ls^2+L_Ls+R_L}{C_LL_Ls^2+1} \right)$$

$$H(s) = \frac{C_LL_3L_LR_Ls^3 + L_3L_Ls^2 + L_3R_Ls}{C_3C_LL_3L_LR_Ls^4 + R_L + s^3(C_3L_3L_L + C_LL_3L_L) + s^2(C_3L_3R_L + C_LL_LR_L) + s(L_3 + L_L)}$$

$$\mathbf{9.45 \quad X-INVALID-ORDER-45} \quad Z(s) = \left(\infty, \infty, \frac{L_3s}{C_3L_3s^2+1}, \infty, \infty, \frac{R_L(C_LL_Ls^2+1)}{C_LL_Ls^2+C_LR_Ls+1} \right)$$

$$H(s) = \frac{C_LL_3L_LR_Ls^3 + L_3R_Ls}{C_3C_LL_3L_LR_Ls^4 + C_LL_3L_Ls^3 + L_3s + R_L + s^2(C_3L_3R_L + C_LL_3R_L + C_LL_LR_L)}$$

$$\mathbf{9.46 \quad X-INVALID-ORDER-46} \quad Z(s) = \left(\infty, \infty, L_3s + R_3 + \frac{1}{C_3s}, \infty, \infty, \frac{1}{C_Ls} \right)$$

$$H(s) = \frac{C_3L_3s^2 + C_3R_3s + 1}{C_3C_LL_3s^3 + C_3C_LR_3s^2 + s(C_3 + C_L)}$$

$$\mathbf{9.47 \quad X-INVALID-ORDER-47} \quad Z(s) = \left(\infty, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_3 L_3 R_L s^2 + C_3 R_3 R_L s + R_L}{C_3 C_L L_3 R_L s^3 + s^2 (C_3 C_L R_3 R_L + C_3 L_3) + s (C_3 R_3 + C_3 R_L + C_L R_L) + 1}$$

$$\mathbf{9.48 \quad X-INVALID-ORDER-48} \quad Z(s) = \left(\infty, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L L_3 R_L s^3 + s^2 (C_3 C_L R_3 R_L + C_3 L_3) + s (C_3 R_3 + C_L R_L) + 1}{C_3 C_L L_3 s^3 + s^2 (C_3 C_L R_3 + C_3 C_L R_L) + s (C_3 + C_L)}$$

$$\mathbf{9.49 \quad X-INVALID-ORDER-49} \quad Z(s) = \left(\infty, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L L_3 L_L s^4 + C_3 C_L L_L R_3 s^3 + C_3 R_3 s + s^2 (C_3 L_3 + C_L L_L) + 1}{C_3 C_L R_3 s^2 + s^3 (C_3 C_L L_3 + C_3 C_L L_L) + s (C_3 + C_L)}$$

$$\mathbf{9.50 \quad X-INVALID-ORDER-50} \quad Z(s) = \left(\infty, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_3 L_3 L_L s^3 + C_3 L_L R_3 s^2 + L_L s}{C_3 C_L L_3 L_L s^4 + C_3 C_L L_L R_3 s^3 + C_3 R_3 s + s^2 (C_3 L_3 + C_3 L_L + C_L L_L) + 1}$$

$$\mathbf{9.51 \quad X-INVALID-ORDER-51} \quad Z(s) = \left(\infty, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L L_3 L_L s^4 + s^3 (C_3 C_L L_3 R_L + C_3 C_L L_L R_3) + s^2 (C_3 C_L R_3 R_L + C_3 L_3 + C_L L_L) + s (C_3 R_3 + C_L R_L) + 1}{s^3 (C_3 C_L L_3 + C_3 C_L L_L) + s^2 (C_3 C_L R_3 + C_3 C_L R_L) + s (C_3 + C_L)}$$

$$\mathbf{9.52 \quad X-INVALID-ORDER-52} \quad Z(s) = \left(\infty, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$$

$$H(s) = \frac{C_3 L_3 L_L R_L s^3 + C_3 L_L R_3 R_L s^2 + L_L R_L s}{C_3 C_L L_3 L_L R_L s^4 + R_L + s^3 (C_3 C_L L_L R_3 R_L + C_3 L_3 L_L) + s^2 (C_3 L_3 R_L + C_3 L_L R_3 + C_3 L_L R_L + C_L L_L R_L) + s (C_3 R_3 R_L + L_L)}$$

$$\mathbf{9.53 \quad X-INVALID-ORDER-53} \quad Z(s) = \left(\infty, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_3 C_L L_3 L_L R_L s^4 + R_L + s^3 (C_3 C_L L_L R_3 R_L + C_3 L_3 L_L) + s^2 (C_3 L_3 R_L + C_3 L_L R_3 + C_L L_L R_L) + s (C_3 R_3 R_L + L_L)}{C_3 C_L L_3 L_L s^4 + s^3 (C_3 C_L L_L R_3 + C_3 C_L L_L R_L) + s^2 (C_3 L_3 + C_3 L_L + C_L L_L) + s (C_3 R_3 + C_3 R_L) + 1}$$

$$\mathbf{9.54 \quad X-INVALID-ORDER-54} \quad Z(s) = \left(\infty, \infty, L_3 s + R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{R_L (C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_3 C_L L_3 L_L R_L s^4 + C_3 C_L L_L R_3 R_L s^3 + C_3 R_3 R_L s + R_L + s^2 (C_3 L_3 R_L + C_L L_L R_L)}{C_3 C_L L_3 L_L s^4 + s^3 (C_3 C_L L_3 R_L + C_3 C_L L_L R_3 + C_3 C_L L_L R_L) + s^2 (C_3 C_L R_3 R_L + C_3 L_3 + C_L L_L) + s (C_3 R_3 + C_3 R_L + C_L R_L) + 1}$$

$$\mathbf{9.55 \quad X-INVALID-ORDER-55} \quad Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L L_3 R_3 R_L s^2 + L_3 R_3 s}{C_3 C_L L_3 R_3 R_L s^3 + R_3 + s^2 (C_3 L_3 R_3 + C_L L_3 R_3 + C_L L_3 R_L) + s (C_L R_3 R_L + L_3)}$$

$$\mathbf{9.56 \quad X-INVALID-ORDER-56} \quad Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L L_3 L_L R_3 s^3 + L_3 R_3 s}{C_3 C_L L_3 L_L R_3 s^4 + C_L L_3 L_L s^3 + L_3 s + R_3 + s^2 (C_3 L_3 R_3 + C_L L_3 R_3 + C_L L_L R_3)}$$

$$\mathbf{9.57 \quad X-INVALID-ORDER-57} \quad Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L L_3 L_L R_3 s^3 + C_L L_3 R_3 R_L s^2 + L_3 R_3 s}{C_3 C_L L_3 L_L R_3 s^4 + R_3 + s^3 (C_3 C_L L_3 R_3 R_L + C_L L_3 L_L) + s^2 (C_3 L_3 R_3 + C_L L_3 R_3 + C_L L_3 R_L + C_L L_L R_3) + s (C_L R_3 R_L + L_3)}$$

$$\mathbf{9.58 \quad X-INVALID-ORDER-58} \quad Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_L L_3 L_L R_3 R_L s^3 + L_3 L_L R_3 s^2 + L_3 R_3 R_L s}{C_3 C_L L_3 L_L R_3 R_L s^4 + R_3 R_L + s^3 (C_3 L_3 L_L R_3 + C_L L_3 L_L R_3 + C_L L_3 L_L R_L) + s^2 (C_3 L_3 R_3 R_L + C_L L_L R_3 R_L + L_3 L_L) + s (L_3 R_3 + L_3 R_L + L_L R_3)}$$

$$\mathbf{9.59 \quad X-INVALID-ORDER-59} \quad Z(s) = \left(\infty, \infty, \frac{L_3 R_3 s}{C_3 L_3 R_3 s^2 + L_3 s + R_3}, \infty, \infty, \frac{R_L (C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_L L_3 L_L R_3 R_L s^3 + L_3 R_3 R_L s}{C_3 C_L L_3 L_L R_3 R_L s^4 + R_3 R_L + s^3 (C_L L_3 L_L R_3 + C_L L_3 L_L R_L) + s^2 (C_3 L_3 R_3 R_L + C_L L_3 R_3 R_L + C_L L_L R_3 R_L) + s (L_3 R_3 + L_3 R_L)}$$

$$\mathbf{9.60 \quad X-INVALID-ORDER-60} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 C_L L_3 R_3 s^3 + C_L R_3 s + s^2 (C_3 L_3 + C_L L_3) + 1}$$

$$\mathbf{9.61 \quad X-INVALID-ORDER-61} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_3 L_3 R_3 R_L s^2 + L_3 R_L s + R_3 R_L}{C_3 C_L L_3 R_3 R_L s^3 + R_3 + R_L + s^2 (C_3 L_3 R_3 + C_3 L_3 R_L + C_L L_3 R_L) + s (C_L R_3 R_L + L_3)}$$

$$\mathbf{9.62 \quad X-INVALID-ORDER-62} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L L_3 R_3 R_L s^3 + R_3 + s^2 (C_3 L_3 R_3 + C_L L_3 R_L) + s (C_L R_3 R_L + L_3)}{s^3 (C_3 C_L L_3 R_3 + C_3 C_L L_3 R_L) + s^2 (C_3 L_3 + C_L L_3) + s (C_L R_3 + C_L R_L) + 1}$$

$$\mathbf{9.63 \quad X-INVALID-ORDER-63} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L L_3 L_L R_3 s^4 + C_L L_3 L_L s^3 + L_3 s + R_3 + s^2 (C_3 L_3 R_3 + C_L L_L R_3)}{C_3 C_L L_3 L_L s^4 + C_3 C_L L_3 R_3 s^3 + C_L R_3 s + s^2 (C_3 L_3 + C_L L_3 + C_L L_L) + 1}$$

$$\mathbf{9.64 \quad X-INVALID-ORDER-64} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_3 L_3 L_L R_3 s^3 + L_3 L_L s^2 + L_L R_3 s}{C_3 C_L L_3 L_L R_3 s^4 + R_3 + s^3 (C_3 L_3 L_L + C_L L_3 L_L) + s^2 (C_3 L_3 R_3 + C_L L_L R_3) + s (L_3 + L_L)}$$

$$\mathbf{9.65 \quad X-INVALID-ORDER-65} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L L_3 L_L R_3 s^4 + R_3 + s^3 (C_3 C_L L_3 R_3 R_L + C_L L_3 L_L) + s^2 (C_3 L_3 R_3 + C_L L_3 R_L + C_L L_L R_3) + s (C_L R_3 R_L + L_3)}{C_3 C_L L_3 L_L s^4 + s^3 (C_3 C_L L_3 R_3 + C_3 C_L L_3 R_L) + s^2 (C_3 L_3 + C_L L_3 + C_L L_L) + s (C_L R_3 + C_L R_L) + 1}$$

$$\mathbf{9.66 \quad X-INVALID-ORDER-66} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$$

$$H(s) = \frac{C_3 L_3 L_L R_3 R_L s^3 + L_3 L_L R_L s^2 + L_L R_3 R_L s}{C_3 C_L L_3 L_L R_3 R_L s^4 + R_3 R_L + s^3 (C_3 L_3 L_L R_3 + C_3 L_3 L_L R_L + C_L L_3 L_L R_L) + s^2 (C_3 L_3 R_3 R_L + C_L L_L R_3 R_L + L_3 L_L) + s (L_3 R_L + L_L R_3 + L_L R_L)}$$

$$\mathbf{9.67 \quad X-INVALID-ORDER-67} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_3 C_L L_3 L_L R_3 R_L s^4 + R_3 R_L + s^3 (C_3 L_3 L_L R_3 + C_L L_3 L_L R_L) + s^2 (C_3 L_3 R_3 R_L + C_L L_L R_3 R_L + L_3 L_L) + s (L_3 R_L + L_L R_3)}{R_3 + R_L + s^4 (C_3 C_L L_3 L_L R_3 + C_3 C_L L_3 L_L R_L) + s^3 (C_3 L_3 L_L + C_L L_3 L_L) + s^2 (C_3 L_3 R_3 + C_3 L_3 R_L + C_L L_L R_3 + C_L L_L R_L) + s (L_3 + L_L)}$$

$$\mathbf{9.68 \quad X-INVALID-ORDER-68} \quad Z(s) = \left(\infty, \infty, \frac{C_3 L_3 R_3 s^2 + L_3 s + R_3}{C_3 L_3 s^2 + 1}, \infty, \infty, \frac{R_L (C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_3 C_L L_3 L_L R_3 R_L s^4 + C_L L_3 L_L R_L s^3 + L_3 R_L s + R_3 R_L + s^2 (C_3 L_3 R_3 R_L + C_L L_L R_3 R_L)}{R_3 + R_L + s^4 (C_3 C_L L_3 L_L R_3 + C_3 C_L L_3 L_L R_L) + s^3 (C_3 C_L L_3 R_3 R_L + C_L L_3 L_L) + s^2 (C_3 L_3 R_3 + C_3 L_3 R_L + C_L L_3 R_L + C_L L_L R_3 + C_L L_L R_L) + s (C_L R_3 R_L + L_3)}$$

$$\mathbf{9.69 \quad X-INVALID-ORDER-69} \quad Z(s) = \left(\infty, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 L_3 R_3 s^2 + R_3}{C_3 C_L L_3 R_3 s^3 + C_3 L_3 s^2 + s (C_3 R_3 + C_L R_3) + 1}$$

$$\mathbf{9.70 \quad X-INVALID-ORDER-70} \quad Z(s) = \left(\infty, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_3 L_3 R_3 R_L s^2 + R_3 R_L}{C_3 C_L L_3 R_3 R_L s^3 + R_3 + R_L + s^2 (C_3 L_3 R_3 + C_3 L_3 R_L) + s (C_3 R_3 R_L + C_L R_3 R_L)}$$

$$\mathbf{9.71 \quad X-INVALID-ORDER-71} \quad Z(s) = \left(\infty, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L L_3 R_3 R_L s^3 + C_3 L_3 R_3 s^2 + C_L R_3 R_L s + R_3}{s^3 (C_3 C_L L_3 R_3 + C_3 C_L L_3 R_L) + s^2 (C_3 C_L R_3 R_L + C_3 L_3) + s (C_3 R_3 + C_L R_3 + C_L R_L) + 1}$$

$$\mathbf{9.72 \quad X-INVALID-ORDER-72} \quad Z(s) = \left(\infty, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_3 C_L L_3 L_L R_3 s^4 + R_3 + s^2 (C_3 L_3 R_3 + C_L L_L R_3)}{C_3 C_L L_3 L_L s^4 + s^3 (C_3 C_L L_3 R_3 + C_3 C_L L_L R_3) + s^2 (C_3 L_3 + C_L L_L) + s (C_3 R_3 + C_L R_3) + 1}$$

$$\mathbf{9.73 \quad X-INVALID-ORDER-73} \quad Z(s) = \left(\infty, \infty, \frac{R_3 (C_3 L_3 s^2 + 1)}{C_3 L_3 s^2 + C_3 R_3 s + 1}, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_3 L_3 L_L R_3 s^3 + L_L R_3 s}{C_3 C_L L_3 L_L R_3 s^4 + C_3 L_3 L_L s^3 + L_L s + R_3 + s^2 (C_3 L_3 R_3 + C_3 L_L R_3 + C_L L_L R_3)}$$

9.74 X-INVALID-ORDER-74 $Z(s) = \left(\infty, \infty, \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \infty, \infty, L_Ls + R_L + \frac{1}{C_Ls} \right)$

$$H(s) = \frac{C_3C_LL_3L_LR_3s^4 + C_3C_LL_3R_3R_Ls^3 + C_LR_3R_Ls + R_3 + s^2(C_3L_3R_3 + C_LL_LR_3)}{C_3C_LL_3L_LR_3s^4 + s^3(C_3C_LL_3R_3 + C_3C_LL_3R_L + C_3C_LL_LR_3) + s^2(C_3C_LR_3R_L + C_3L_3 + C_LL_L) + s(C_3R_3 + C_LR_3 + C_LR_L) + 1}$$

9.75 X-INVALID-ORDER-75 $Z(s) = \left(\infty, \infty, \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \infty, \infty, \frac{L_LR_Ls}{C_LL_LR_Ls^2+L_Ls+R_L} \right)$

$$H(s) = \frac{C_3L_3L_LR_3R_Ls^3 + L_LR_3R_Ls}{C_3C_LL_3L_LR_3R_Ls^4 + R_3R_L + s^3(C_3L_3L_LR_3 + C_3L_3L_LR_L) + s^2(C_3L_3R_3R_L + C_3L_LR_3R_L + C_LL_LR_3R_L) + s(L_LR_3 + L_LR_L)}$$

9.76 X-INVALID-ORDER-76 $Z(s) = \left(\infty, \infty, \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \infty, \infty, \frac{C_LL_LR_Ls^2+L_Ls+R_L}{C_LL_Ls^2+1} \right)$

$$H(s) = \frac{C_3C_LL_3L_LR_3R_Ls^4 + C_3L_3L_LR_3s^3 + L_LR_3s + R_3R_L + s^2(C_3L_3R_3R_L + C_LL_LR_3R_L)}{R_3 + R_L + s^4(C_3C_LL_3L_LR_3 + C_3C_LL_3L_LR_L) + s^3(C_3C_LL_LR_3R_L + C_3L_3L_L) + s^2(C_3L_3R_3 + C_3L_3R_L + C_3L_LR_3 + C_LL_LR_3 + C_LL_LR_L) + s(C_3R_3R_L + L_L)}$$

9.77 X-INVALID-ORDER-77 $Z(s) = \left(\infty, \infty, \frac{R_3(C_3L_3s^2+1)}{C_3L_3s^2+C_3R_3s+1}, \infty, \infty, \frac{R_L(C_LL_Ls^2+1)}{C_LL_Ls^2+C_LR_Ls+1} \right)$

$$H(s) = \frac{C_3C_LL_3L_LR_3R_Ls^4 + R_3R_L + s^2(C_3L_3R_3R_L + C_LL_LR_3R_L)}{R_3 + R_L + s^4(C_3C_LL_3L_LR_3 + C_3C_LL_3L_LR_L) + s^3(C_3C_LL_3R_3R_L + C_3C_LL_LR_3R_L) + s^2(C_3L_3R_3 + C_3L_3R_L + C_LL_LR_3 + C_LL_LR_L) + s(C_3R_3R_L + C_LR_3R_L)}$$

10 X-INVALID-WZ

11 X-PolynomialError